DEFINITE INTEGRALS TEST
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Higher 2

MATHEMATICS 9758

Definite Integrals **2026**
QUESTION PAPER **1 hour 45 minutes**

Additional Materials: Writing paper
List of Formulae and Results (MF27)

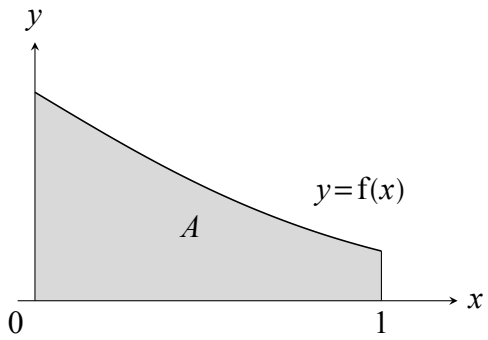
READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 58.

1 The diagram shows a sketch of the curve $y = \mathrm{f}(x)$. The shaded region under the curve between $x = 0$ and $x = 1$, as shown in the diagram, is A. This region is split into n vertical strips of equal width, h.



(a) State the value of h. [1]

(b) Hence show, using a sketch, that $\sum_{r=1}^{n} h\mathrm{f}(rh)$ is less than the area of A. [2]

You are now given that $\mathrm{f}(x) = \dfrac{\mathrm{e}^{-2x}}{1+\mathrm{e}^{-2x}}$.

(c) Hence find the exact value of $\lim_{n\to\infty} \sum_{r=1}^{n} h\mathrm{f}(rh)$. [3]

[ACJC/2026 Prelim/P1]

2 A curve C has parametric equations

$$x=-\sqrt{t^2+4}, \quad y=\frac{\ln t}{t}, \quad \text{where } t>0.$$

(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{(\ln t-1)\sqrt{t^2+4}}{t^3}$. [3]

(b) It is given that the area under a curve defined parametrically by $x=\mathrm{f}(t), y=\mathrm{g}(t)$ is

$$\int_a^b y\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t.$$

Show that the area bounded by C and the lines $x=-\sqrt{13}$ and $x=-\sqrt{5}$ is given by

$$\int_1^3 \frac{\ln t}{\sqrt{t^2+4}}\,\mathrm{d}t.$$

Find the area, giving your answer to 4 decimal places. [3]

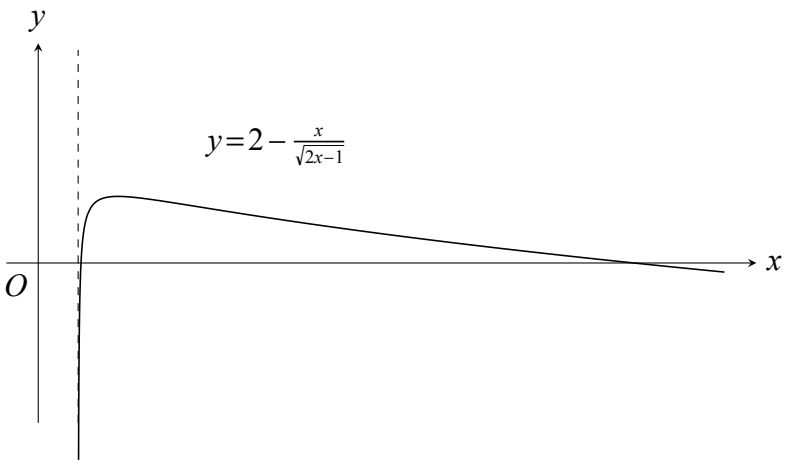
(c) It is given that the length of the arc of a curve defined parametrically by $x=\mathrm{f}(t)$, $y=\mathrm{g}(t)$, from $t=t_1$ to $t=t_2$, is $\displaystyle\int_{t_1}^{t_2}\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$.

Find the length of the arc of C from $t=1$ to $t=3$, giving your answer to 3 significant figures. [3]

[TMJC/2019 Prelim/P2 (Modified)]

3 **(a)** Using the substitution $u=\sqrt{2x-1}$, show that $\int \frac{x}{\sqrt{2x-1}}\,\mathrm{d}x=\int \frac{u^2+1}{2}\,\mathrm{d}u$. [2]

(b) **(i)** The diagram below shows the graph of $y=2-\frac{x}{\sqrt{2x-1}}$. Show, on a copy of this diagram, the equation of the asymptote of the curve and the coordinates of its turning point and points of intersection with the x-axis. [2]



(ii) The region R is bounded by the curve $y=2-\frac{x}{\sqrt{2x-1}}$ and the lines $x=1$, $x=3$ and $y=\frac{1}{2}$. Find the exact area of R. [5]

(c) The region S is bounded by the curves $y=2-\frac{x}{\sqrt{2x-1}}$, $x=2(y-1)^2+1$, the lines $x=1$, $x=4$ and the x-axis.

(i) Express y in terms of x for the curve $x=2(y-1)^2+1$, giving both branches. [1]

(ii) On the same diagram as in part **(b)(i)**, sketch the graph of $x=2(y-1)^2+1$. [2]

(iii) Find the volume of the solid generated when S is rotated through 2π radians about the x-axis. Give your answer correct to 3 decimal places. [3]

[RI/2025 Prelim/P1]

4 In this question $\sin^{-1}(1-x)$ refers to the principal value.

(a) **(i)** Use the substitution $z = \sin^{-1}(1-x)$ to show that $\int_0^1 \sin^{-1}(1-x)\,\mathrm{d}x = \int_a^b z\cos z\,\mathrm{d}z$, where a and b are constants to be determined. [2]

(ii) Hence show that $\int_0^1 \sin^{-1}(1-x)\,\mathrm{d}x = \frac{\pi}{2} - 1$. [2]

(b) A curve C is given by the equation

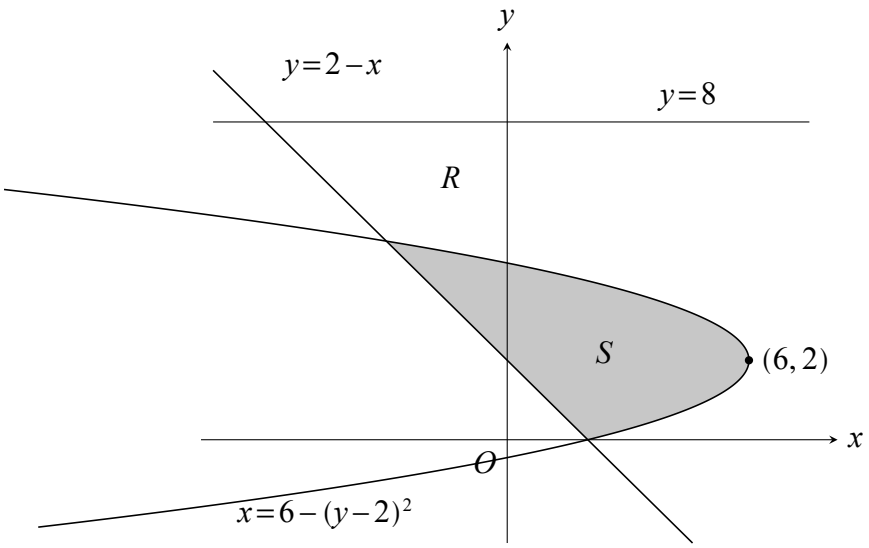
$$y = \sin^{-1}(1-x) \quad \text{for } 0 \le x \le 2.$$

(i) Find exactly $\int_0^2 \left|\sin^{-1}(1-x)\right|\,\mathrm{d}x$. [2]

(ii) The region R is bounded by the curve C, the line $x=2$ and the part of the x-axis between 1 and 2. Find the exact volume generated when R is rotated 360° about the y-axis. [4]

[ACJC/2026 Prelim/P2]

5 In the diagram below, the region R is bounded by the curve C with equation $x = 6-(y-2)^2$, the lines $y=8$, $y=2-x$ and the y-axis. The region S is bounded by C and the line $y=2-x$.



(a) Find the exact area of region R. [5]

(b) Find the volume of the solid of revolution formed when region S is rotated through 360° about the x-axis, leaving your answer to 2 decimal places. [3]

[VJC/2025 Prelim/P1]

6 A curve C has equation $y = xe^{-x}$, $x \geq 0$.

(a) Show that C has exactly one stationary point. [2]

(b) Sketch C, giving the exact coordinates of the stationary point. [1]

The region R is bounded by C, the y-axis and the line $y = \frac{1}{e}$.

(c) Find the exact area of R. [4]

(d) An ornament is formed by rotating R through 2π radians about the line $y = 2$. By considering a suitable translation of C, find the volume of the ornament. [3]

[RI/2026 Prelim/P1]